

Definition 1 The universal quantification of $P(x)$ is the statement
 " $P(x)$ for all values of x in the domain."

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here, $\forall$ is called the universal quantifier. We read $\forall x P(x)$ as "for all x $P(x)$ " or "for every x $P(x)$ ". An element for which $P(x)$ is false is called a counter example to $\forall x P(x)$.

Example 8: Let $P(x)$ be the statement " $x+1 > x$ ". What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Soln: Let, $\exists x$ s.t. $x+1 \leq x \Rightarrow 1 \leq 0$. This is False.
 $\therefore$ For all x , $x+1 > x$ must be True.

Table 1: Quantifiers

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is False
$\exists x P(x)$	There is an x for which $P(x)$ is true	$P(x)$ is false for every x

Example 9: Let $Q(x)$ be the statement " $x < 2$ ". What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Soln: $Q(3): 3 < 2 \rightarrow \text{False}$. $\therefore \forall x Q(x)$ is False

Example 11: What does the statement $\forall x N(x)$ mean if $N(x)$ is "Computer x is connected to the network" and the domain consists of all computers on campus?

Soln: The statement $\forall x N(x)$ means that for every computer x on campus, that computer x is connected to the network. This statement can be expressed in English as "Every computer on campus is connected to the network."